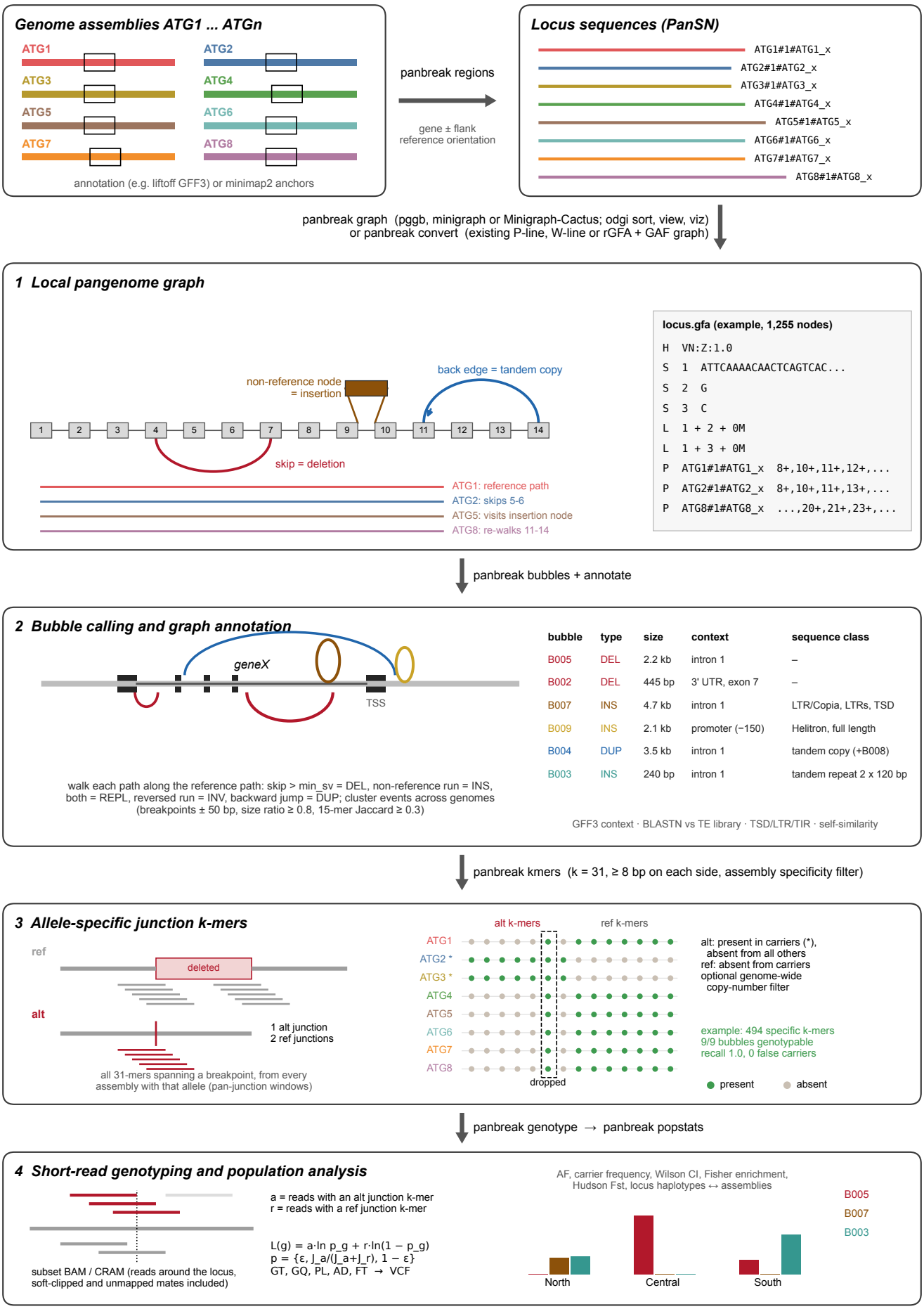

Genome assemblies ATG1 ... ATGn
ATG1
ATG2
ATG3
ATG4
ATG5
ATG6
ATG7
ATG8
annotation (e.g. liftoff GFF3) or minimap2 anchors
panbreak regions
gene ± flank
reference orientation
Locus sequences (PanSN)
ATG1#1#ATG1_x
ATG2#1#ATG2_x
ATG3#1#ATG3_x
ATG4#1#ATG4_x
ATG5#1#ATG5_x
ATG6#1#ATG6_x
ATG7#1#ATG7_x
ATG8#1#ATG8_x
panbreak graph (pggb, minigraph or Minigraph-Cactus; odgi sort, view, viz)
or panbreak convert (existing P-line, W-line or rGFA + GAF graph)
1 Local pangenome graph
non-reference node
= insertion
back edge = tandem copy
1
2
3
4
5
6
7
8
9
10
11
12
13
14
skip = deletion
ATG1: reference path
ATG2: skips 5-6
ATG5: visits insertion node
ATG8: re-walks 11-14
locus.gfa (example, 1,255 nodes)
H VN:Z:1.0
S 1 ATTCAAAACAACTCAGTCAC...
S 2 G
S 3 C
L 1 + 2 + 0M
L 1 + 3 + 0M
P ATG1#1#ATG1_x 8+,10+,11+,12+,...
P ATG2#1#ATG2_x 8+,10+,11+,13+,...
P ATG8#1#ATG8_x ...,20+,21+,23+,...
panbreak bubbles + annotate
2 Bubble calling and graph annotation
geneX
TSS
walk each path along the reference path: skip > min_sv = DEL, non-reference run = INS,
both = REPL, reversed run = INV, backward jump = DUP; cluster events across genomes
(breakpoints ± 50 bp, size ratio ≥ 0.8, 15-mer Jaccard ≥ 0.3)
bubble type size context sequence class
B005 DEL 2.2 kb intron 1 –
B002 DEL 445 bp 3' UTR, exon 7 –
B007 INS 4.7 kb intron 1 LTR/Copia, LTRs, TSD
B009 INS 2.1 kb promoter (−150) Helitron, full length
B004 DUP 3.5 kb intron 1 tandem copy (+B008)
B003 INS 240 bp intron 1 tandem repeat 2 x 120 bp
GFF3 context · BLASTN vs TE library · TSD/LTR/TIR · self-similarity
panbreak kmers (k = 31, ≥ 8 bp on each side, assembly specificity filter)
3 Allele-specific junction k-mers
ref
deleted
alt
1 alt junction
2 ref junctions
all 31-mers spanning a breakpoint, from every
assembly with that allele (pan-junction windows)
alt k-mers
ref k-mers
ATG1
ATG2 *
ATG3 *
ATG4
ATG5
ATG6
ATG7
ATG8
dropped
alt: present in carriers (*),
absent from all others
ref: absent from carriers
optional genome-wide
copy-number filter
example: 494 specific k-mers
9/9 bubbles genotypable
recall 1.0, 0 false carriers
present
absent
panbreak genotype → panbreak popstats
4 Short-read genotyping and population analysis
subset BAM / CRAM (reads around the locus,
soft-clipped and unmapped mates included)
a = reads with an alt junction k-mer
r = reads with a ref junction k-mer
L(g) = a·ln p_g + r·ln(1 − p_g)
p = {ε, J_a/(J_a+J_r), 1 − ε}
GT, GQ, PL, AD, FT → VCF
AF, carrier frequency, Wilson CI, Fisher enrichment,
Hudson Fst, locus haplotypes ↔ assemblies
B005
B007
B003
North
Central
South